

Joint density function

1. The jpdf of (X,Y) is given by $f(x,y)=e^{-(x+y)}, 0 \leq x,y \leq \infty$.

Are X and Y are independent?

The marginal density function of X is given by

$$f(x) = \int_0^{\infty} e^{-(x+y)} dy = e^{-x} \left[-e^{-y} \right]_0^{\infty} = e^{-x}, 0 \leq x \leq \infty$$

The marginal density function of Y is given by

$$f(y) = \int_0^{\infty} e^{-(x+y)} dx = e^{-y} \left[-e^{-x} \right]_0^{\infty} = e^{-y}, 0 \leq y \leq \infty$$

$$f(x, y) = f(x) \cdot f(y)$$

$$e^{-(x+y)} = e^{-x} e^{-y} \Rightarrow e^{-(x+y)}$$

$\therefore$ X and Y are independent.

2. If the joint density function of the two random variables X and Y be

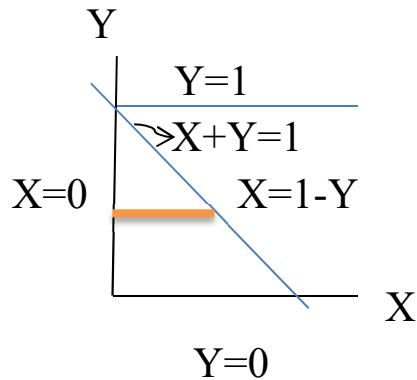
$$f(x, y) = \begin{cases} e^{-(x+y)}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Find (i) $P(X < 1)$ (ii) $P(X+Y < 1)$

$$\text{i) } P(X < 1) = \int_0^1 f(x) dx = \int_0^1 e^{-x} dx = 1 - e^{-1}$$

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$$\text{ii) } P(X+Y < 1) = \int_0^1 \int_0^{1-y} e^{-(x+y)} dx dy = \int_0^1 e^{-y} \left[-e^{-x} \right]_0^{1-y} dy = -[e^{-1} + e^{-1} - 1] = 1 - 2e^{-1}$$



3. Suppose the pdf of X, Y is

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2), & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Obtain the marginal pdf of X and Y.

Find $P \left[\frac{1}{4} \leq y \leq \frac{3}{4} \right]$

Marginal pdf of X:

$$f(x) = \int_0^1 f(x, y) dy = \int_0^1 \frac{6}{5}(x + y^2) dy = \frac{6}{5} \left(x + \frac{1}{3} \right)$$

Marginal Pdf of Y

$$f(y) = \int_0^1 f(x, y) dx = \int_0^1 \frac{6}{5}(x + y^2) dx = \frac{6}{5} \left[\frac{1}{2} + y^2 \right]$$

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$$P\left(\frac{1}{4} \leq Y \leq \frac{3}{4}\right) = \int_{1/4}^{3/4} f(y) dy = \int_{1/4}^{3/4} \frac{6}{5} \left(\frac{1}{2} + y^2\right) dy = 37/80$$

Practice problems:

4. X and Y are two random variable with joint pdf

$$f(x,y) = \frac{1}{8}(6-x-y), 0 < x < 2, 2 < y < 4$$

Find 1) $P(X < 1 \cap Y < 3)$ 2) $P(X < 1/Y < 3)$

$$\text{Hint: 1) } P(X < 1 \cap Y < 3) = \int_0^1 \int_2^3 f(x,y) dy dx = 3/8$$

$$2) P(X < 1/Y < 3) = \frac{P(X < 1 \cap Y < 3)}{P(Y < 3)}$$

$$P(Y < 3) = \int_2^3 f(y) dy = \int_2^3 \frac{1}{8}(10-2y) dy = 5/8$$

(Marginal probability density function of y=

$$f(y) = \int_0^2 f(x,y) dx = \frac{1}{8}(10-2y)$$

$$P(X < 1/Y < 3) = \frac{P(X < 1 \cap Y < 3)}{P(Y < 3)} = \frac{3/8}{5/8} = \frac{3}{5}$$

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5. The Joint probability mass function of X and Y is

X \ Y	0	1	2
0	0.1	0.04	0.02
1	0.08	0.2	0.06
2	0.06	0.14	0.3

1) Compute the marginal pmf of X and Y.

2) $P[X \leq 1, Y \leq 1]$

3) Check if X and Y are independent.

$$1) P(X)=0.16 \quad 0.34 \quad 0.5 \quad P(Y)=0.24 \quad 0.38 \quad 0.38$$

$$\begin{aligned}
 2) \quad P[X \leq 1, Y \leq 1] &= P[X \leq 1, Y=0] + P[X \leq 1, Y=1] \\
 &= P[X=0, Y=0] + P[X=1, Y=0] + P[X=0, Y=1] + P[X=1, Y=1] \\
 &= 0.42
 \end{aligned}$$

3) To test whether X and Y are independent:

$$P[X=0] \cdot P[Y=0] = 0.16 \cdot 0.24 \neq 1$$

$$P[X=0] \cdot P[Y=0] \neq P[X=0, Y=0]$$

$\Rightarrow$ X and Y are not independent.

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